
Statistical Inference

B. Statistical Data Science 2nd Year Indian Statistical Institute

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Exercise Series 14

Exercise 1. For the following examples, find a $100(1 - \alpha)\%$ confidence interval/set by inverting the UMPU test. Does inverting the UMP test give you a one-sided interval? Also construct the $100(1 - \alpha)\%$ confidence intervals/sets by inverting the likelihood ratio test, the Wald's test and the Rao's score test. Compare the coverage probabilities of these different intervals/sets.

- (a) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Binomial}(p), 0 \leq p \leq 1. \theta = p.$
- (b) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Poisson}(\lambda), \lambda > 0. \theta = \lambda$
- (c) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Geometric}(p), 0 \leq p \leq 1. \theta = p.$
- (d) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Uniform}(0, \theta), \theta > 0.$
- (e) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(\mu, 1), \mu \in \mathbb{R}. \theta = \mu.$
- (f) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(0, \sigma^2), \sigma > 0.$
- (g) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Exponential}(\theta), \theta > 0.$
- (h) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Laplace}(\mu, 1), \mu \in \mathbb{R}. \theta = \mu.$
- (i) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Laplace}(0, \sigma), \sigma > 0. \theta = \sigma.$

Exercise 2. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(p_1)$ and $Y_1, \dots, Y_m \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(p_2)$ be two independent samples. Construct $100(1 - \alpha)\%$ confidence intervals/sets for $p_1 - p_2$ by inverting the Wald's test, Rao's score test and likelihood ratio test. How would you construct a $100(1 - \alpha)\%$ upper confidence interval (of the form $[a, \infty)$) for $p_1 - p_2$?

Exercise 3. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(\mu_1, \sigma^2)$ and $Y_1, \dots, Y_m \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(\mu_2, \sigma^2)$ be two independent samples, where $\mu_1 \in \mathbb{R}, \mu_2 \in \mathbb{R}$ and $\sigma^2 > 0$ are unknown parameters. Construct $100(1 - \alpha)\%$ confidence intervals/set for $\mu_1 - \mu_2$ by inverting the Wald's test, Rao's score test and likelihood ratio test. Construct a $100(1 - \alpha)\%$ confidence interval for $\mu_1 - \mu_2$ of the form $(-\infty, b]$.

Exercise 4. Let $(X_1, Y_1), \dots, (X_n, Y_n) \stackrel{\text{i.i.d.}}{\sim} \text{BVN}(\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho)$, where $\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho$ are all unknown. Construct a $100(1 - \alpha)\%$ confidence intervals/sets for $\mu_x - \mu_y$.

Exercise 5. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Geometric}(p_1)$ and $Y_1, \dots, Y_m \stackrel{\text{i.i.d.}}{\sim} \text{Geometric}(p_2)$ be two independent samples. Construct $100(1 - \alpha)\%$ confidence interval for $p_1 - p_2$ by inverting the likelihood ratio test, Wald's test and Rao's score test.

Exercise 6. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(\mu, 1)$. Consider the problem of testing $\mathcal{H}_0 : \mu = 0$ vs. $\mathcal{H}_a : \mu > 0$. We reject $\mathcal{H}_0$ if the value of $\bar{X}$ is large (i.e., $\bar{X} > c$). The (right-tailed) p -value for this test is defined as

$$p_{\text{right}} = p_{\text{right}}(\mathbf{x}) = \mathbb{P}_{\mathcal{H}_0}(\bar{X} > \bar{x}_{\text{obs}}),$$

where $\bar{x}_{\text{obs}}$ is the mean of the observed sample $\mathbf{x}$.

- (a) Find a closed-form expression for $p_{\text{right}}(\mathbf{x})$.
- (b) Show that under $\mathcal{H}_0$, $p_{\text{right}}(\mathbf{X}) \sim \text{Uniform}(0, 1)$.

We can also define the left-tailed/both-tailed p -values as

$$p_{\text{left}} = p_{\text{left}}(\mathbf{x}) = \mathbb{P}_{\mathcal{H}_0}(\bar{X} < \bar{x}_{\text{obs}}) \quad \text{and} \quad p_{\text{both}} = p_{\text{both}}(\mathbf{x}) = \mathbb{P}_{\mathcal{H}_0}(|\bar{X}| > |\bar{x}_{\text{obs}}|).$$

- (c) Find closed-form expressions for $p_{\text{left}}(\mathbf{x})$ and $p_{\text{both}}(\mathbf{x})$.
- (d) Show that under $\mathcal{H}_0$, $p_{\text{left}} \sim \text{Uniform}(0, 1)$ and $p_{\text{both}} \sim \text{Uniform}(0, 1)$.

Exercise 7. In general, consider the problem of testing $\mathcal{H}_0 : \theta = \theta_0$ against $\mathcal{H}_{\text{right}} : \theta > \theta_0 / \mathcal{H}_{\text{left}} : \theta < \theta_0 / \mathcal{H}_{\text{both}} : \theta \neq \theta_0$. Consider the following tests

$$\varphi_{\text{right}}(\mathbf{x}) = \mathbf{1}\{T(\mathbf{x}) > c_{\text{right}}\}, \quad \varphi_{\text{left}}(\mathbf{x}) = \mathbf{1}\{T(\mathbf{x}) < c_{\text{left}}\}, \quad \varphi_{\text{both}}(\mathbf{x}) = \mathbf{1}\{|T(\mathbf{x})| > c_{\text{both}}\}.$$

Here, $T = T(\mathbf{X})$ is the test statistic. The corresponding p -values are defined as

$$p_{\text{right}}(\mathbf{x}) = \mathbb{P}_{\mathcal{H}_0}(T(\mathbf{X}) > t_{\text{obs}}), \quad p_{\text{left}}(\mathbf{x}) = \mathbb{P}_{\mathcal{H}_0}(T(\mathbf{X}) < t_{\text{obs}}), \quad p_{\text{both}}(\mathbf{x}) = \mathbb{P}_{\mathcal{H}_0}(|T(\mathbf{X})| > |t_{\text{obs}}|),$$

where $t_{\text{obs}} = T(\mathbf{x})$ is the observed value of the test statistic. Suppose that, under $\mathcal{H}_0$, the distribution of T is continuous. Verify that, under $\mathcal{H}_0$, $p_{\text{right}}(\mathbf{X}), p_{\text{left}}(\mathbf{X}), p_{\text{both}}(\mathbf{X})$ have $\text{Uniform}(0, 1)$ distribution.